Derivation of Roe Flux for Shallow Water Equations

The 1D shallow water equations are:

$$\mathbf{u} = \begin{bmatrix} h \\ hu \end{bmatrix}, \quad \mathbf{f} = \begin{bmatrix} hu \\ hu^2 + \frac{1}{2}gh^2 \end{bmatrix}$$

where h is water height, u is velocity, hu is momentum, and g is gravitational acceleration.

Roe Flux Process

1. **Formulate the Roe Flux**: The numerical flux is:

$$\mathbf{F}_{i+1/2} = \frac{1}{2}\left(\mathbf{f}(\mathbf{u}_L) + \mathbf{f}(\mathbf{u}_R)\right) - \frac{1}{2}\sum_{k=1}^{2} |\tilde{\lambda}_k| \tilde{\alpha}_k \tilde{\mathbf{r}}_k$$

   where $\mathbf{u}_L$ and $\mathbf{u}_R$ are left and right states, $\tilde{\lambda}_k$, $\tilde{\alpha}_k$, and $\tilde{\mathbf{r}}_k$ are the eigenvalues, wave strengths, and eigenvectors of the Roe-averaged Jacobian.

2. **Jacobian**: The flux Jacobian is:

$$\mathbf{A} = \begin{bmatrix} 0 & 1 \\ -u^2 + gh & 2u \end{bmatrix}$$

   Eigenvalues: $\lambda_1 = u - \sqrt{gh}$, $\lambda_2 = u + \sqrt{gh}$. Eigenvectors:

$$\mathbf{r}_1 = \begin{bmatrix} 1 \\ u - \sqrt{gh} \end{bmatrix}, \quad \mathbf{r}_2 = \begin{bmatrix} 1 \\ u + \sqrt{gh} \end{bmatrix}$$

3. **Roe Averaging**: For states $\mathbf{u}_L = [h_L, h_L u_L]^T$ and $\mathbf{u}_R = [h_R, h_R u_R]^T$:

$$\tilde{h} = \sqrt{h_L h_R}, \quad \tilde{u} = \frac{\sqrt{h_L}u_L + \sqrt{h_R}u_R}{\sqrt{h_L} + \sqrt{h_R}}$$

   Roe-averaged eigenvalues: $\tilde{\lambda}_1 = \tilde{u} - \sqrt{g\tilde{h}}$, $\tilde{\lambda}_2 = \tilde{u} + \sqrt{g\tilde{h}}$. Eigenvectors:

$$\tilde{\mathbf{r}}_1 = \begin{bmatrix} 1 \\ \tilde{u} - \sqrt{g\tilde{h}} \end{bmatrix}, \quad \tilde{\mathbf{r}}_2 = \begin{bmatrix} 1 \\ \tilde{u} + \sqrt{g\tilde{h}} \end{bmatrix}$$

4. **Wave Strengths**: Decompose the jump $\Delta\mathbf{u} = \mathbf{u}_R - \mathbf{u}_L = [h_R - h_L, h_R u_R - h_L u_L]^T$:

$$\Delta\mathbf{u} = \tilde{\alpha}_1 \tilde{\mathbf{r}}_1 + \tilde{\alpha}_2 \tilde{\mathbf{r}}_2$$

   Solve:

$$\tilde{\alpha}_1 = \frac{\Delta(hu) - \tilde{u}\Delta h + \sqrt{g\tilde{h}}\Delta h}{2\sqrt{g\tilde{h}}}, \quad \tilde{\alpha}_2 = \frac{\Delta(hu) - \tilde{u}\Delta h - \sqrt{g\tilde{h}}\Delta h}{-2\sqrt{g\tilde{h}}}$$

   where $\Delta h = h_R - h_L$, $\Delta(hu) = h_R u_R - h_L u_L$.

5. **Roe Flux**: Compute:

$$\mathbf{F}_{i+1/2} = \frac{1}{2}\left(\mathbf{f}(\mathbf{u}_L) + \mathbf{f}(\mathbf{u}_R)\right) - \frac{1}{2}\left(|\tilde{\lambda}_1|\tilde{\alpha}_1\tilde{\mathbf{r}}_1 + |\tilde{\lambda}_2|\tilde{\alpha}_2\tilde{\mathbf{r}}_2\right)$$

This ensures a conservative, consistent flux satisfying the Rankine-Hugoniot condition.